

Revised Research Proposal

Label-Free Irreversibility as a Safety Signal in World-Model Latents

The Safety Dial: Deployment-Time Conservatism

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Abstract

Latent safety filters removed the hand-tuned penalty weight from safe reinforcement learning, but they did not remove supervision: the failure set that seeds the reachability computation is a classifier trained on human-annotated failure observations, so the filter cannot see a failure mode nobody labelled. This proposal asks whether that failure set can instead be derived from the dynamics, using *irreversibility* as the criterion, and whether the resulting conservatism can be exposed as a threshold an operator sets at deployment rather than a constant an engineer fixes during training.

The central methodological change from the August 2026 proposal is the evaluation environment. The Safety-Gymnasium locomotion suite carries two distinct unsafe events: a velocity-limit violation, which is extrinsic, recoverable, and is the cost signal every published safe-RL baseline optimises; and falling over, which is intrinsic, irreversible, and is reported through episode termination rather than through the cost channel. These two signals are produced by different mechanisms, which provides a labelled mode and an unlabelled mode inside a single recognised benchmark, and therefore a direct test of whether a label-free method detects an unsafe event that a supervised one structurally cannot. A feasibility probe is reported. The work is scoped to an Honours thesis completing in November 2026 and a workshop submission in February 2027.

1 What changed, and why

The August 2026 proposal committed to label-free latent reachability, with OGBench-Cube as the primary environment and a bespoke annotation harness supplying ground-truth recoverability labels for evaluation. That commitment is retained. Three things change.

Environment. Safety-Gymnasium was previously rejected on the grounds that its hazards are recoverable. That reasoning was correct for the navigation suite, where hazards are regions marked on the floor with an assigned cost and nothing in the dynamics distinguishes them. It does not hold for the locomotion suite, where the robot can fall and cannot get up. Section 3 sets out the distinction and the evidence. Adopting a recognised benchmark supplies published baselines and removes the need to hand-build an annotation set, which was a substantial and bias-prone piece of work.

Scope. The August proposal carried four hypotheses across five stages, and a subsequent internal planning note carried four further hypotheses across six stages. Neither set has been reduced by experiment, and the combination is not deliverable in the time available. This document commits to three hypotheses and states the gate that kills each.

Venue. The target is an ICLR 2027 workshop rather than the main track. The main track abstract deadline of 18 September 2026 passed deliberately without a submission: the direction was not settled and a rushed abstract was not worth the slot. Workshop papers are non-archival and ICLR’s dual-submission policy permits them, so this preserves a later full submission. Section 9 gives the schedule.

2 Background and the gap

Safe reinforcement learning has historically specified unsafe behaviour through a penalty term in the reward or a cost function under a constraint. Both require a human to quantify harm in advance, and the resulting policy is sensitive to that quantity. ROSARL (Nangue Tasse et al. 2023) reduces the problem by deriving a sufficient penalty from intrinsic quantities rather than tuning it, but the operating point remains a single scalar fixed before deployment.

Latent safety filters (Nakamura, Peters, and Bajcsy 2025) take a different route. They compute a Hamilton-Jacobi backward reachable tube (Bansal et al. 2017) in the latent space of a learned world model, and use it to override a task policy when failure becomes inevitable. This removes the penalty weight. UNISafe (Seo, Nakamura, and Bajcsy 2025) adds an uncertainty axis so that the filter distrusts its own rollouts near the boundary, and AnySafe (Agrawal et al. 2025) makes the constraint adjustable at runtime through conformal similarity.

The gap is that every one of these seeds its reachability computation with a *classifier over labelled failure observations*. The filter is therefore blind to any failure mode that was not annotated. Replacing the penalty weight with a labelled failure set relocates the specification problem rather than solving it.

Irreversibility is the natural label-free replacement. A transition is unsafe when no available action sequence returns the system to where it was. This is a property of the transition structure, requires no annotation, and is computable only with a world model, since it depends on rollouts of actions that were never taken. Reversibility-aware reinforcement learning (Grinsztajn et al. 2021) arrives at the same criterion from the alignment side, using a precedence classifier over unlabelled trajectories.

A second gap concerns the substrate. Every latent safety filter published to date runs on a *reconstructive* world model, where the latent is trained to support pixel reconstruction and therefore retains the information a safety signal needs more or less by construction. A joint-embedding predictive architecture (JEPA) is trained to keep what is predictable and discard the rest. Whether a non-reconstructive latent retains what safety needs is an open question, and it is the representation half of this proposal.

3 The key observation

The Safety-Gymnasium locomotion suite (Ji et al. 2023) defines cost as a velocity threshold: cost accrues when the agent’s forward velocity exceeds a fixed limit. Falling over is not part of the cost signal at all. It is reported through the `terminated` flag inherited from the underlying MuJoCo locomotion task, because a fallen Hopper or Walker is in an unhealthy state from which the episode cannot continue.

The two events therefore differ on every axis that matters here:

This yields, inside a single recognised benchmark, exactly the structure the zero-shot transfer experiment requires: a labelled failure mode that the supervised baseline is trained on, and an unlabelled, genuinely irreversible failure mode on which both methods are evaluated. No bespoke annotation harness is needed, because the simulator reports both modes directly and they can be reserved for evaluation.

	Mode A: velocity violation	Mode B: falling
Channel	cost	terminated
Origin	extrinsic, hand-set limit	intrinsic, physical
Reversible	yes, decelerate	no, the robot cannot rise
In cost signal	yes	no
Baselines optimise it	yes	no

Table 1: The two unsafe events in the Safety-Gymnasium locomotion suite.

3.1 Feasibility probe

A short probe was run to confirm the two channels are mechanically distinct. Twenty episodes per condition, two policies: uniform random actions, and a crude directed policy applying maximum forward torque as a proxy for a return-seeking agent.

Environment	Policy	Episodes with fall	Episodes with cost	Mean length
SafetyHopperVelocity-v1	random	20/20	1/20	24.4
SafetyHopperVelocity-v1	full forward	20/20	20/20	21.9
SafetyWalker2dVelocity-v1	random	20/20	0/20	21.5
SafetyWalker2dVelocity-v1	full forward	20/20	0/20	207.1

Table 2: Feasibility probe. Falls are frequent; cost is frequently absent.

Two readings, and the second matters more than the first.

The structural point holds unconditionally: falling and cost are produced by different mechanisms, so a filter seeded from the cost signal has no information about falls. Walker2d under the directed policy is the clearest case, walking for an average of 207 steps, never once triggering cost, and falling in every episode.

The empirical point is weaker and is stated plainly. Hopper under the directed policy triggers cost in every episode as well as falling, so the two signals *co-occur* in that regime and the modes are not cleanly separable there. Disjointness is therefore a property of the robot and the policy, not of the benchmark as a whole. Under a random or a crude policy the robots also fall almost immediately, which is not the regime of interest.

Gate 0. Before any comparison is run, fall rate and cost rate must be measured under *trained* policies (unconstrained PPO and a Lagrangian-constrained variant) on each candidate robot, and the environment selected on measured disjointness. If no robot shows falls at a usable rate under a trained policy while cost remains rare, the transfer experiment is void and the environment choice must be revisited. This mirrors the dataset-composition control in the August proposal, which exists for the same reason.

4 Research question and hypotheses

Research question. Can an unsafe event be detected from irreversibility in the dynamics of a learned world model, without failure labels, when that event is absent from the benchmark’s own cost signal and therefore from every supervised baseline trained on it; and does a non-reconstructive latent retain the information required to do so?

H1 (Representation). Information sufficient to predict an imminent irreversible failure is linearly decodable from world-model latents, and degrades measurably with predictor rollout depth. Frozen general-purpose features retain more of it than an end-to-end

predictive latent, because predictive training discards the fine pose and balance detail that signals an imminent fall.

H2 (Detection). A label-free irreversibility score computed in that latent separates pre-failure states from matched recoverable states at $\text{AUROC} \geq 0.8$, using no failure labels at any point in its construction.

H3 (Transfer and dial). A filter built on H2 reduces the irreversible-failure rate that the cost signal ignores, at a task-return cost selected at deployment by moving a single threshold, recovering a safety-performance frontier that penalty-based baselines require one training run per operating point to trace.

H1 is the representation claim, H2 the estimator, H3 the control claim. H1 and H3 are the two the project is built around; H2 is the bridge between them.

5 Method

Stage 0: harness and environment selection. Collect trajectories from the candidate locomotion environments under random, trained-unconstrained and trained-constrained policies, recording pixels, simulator state, action, cost, termination, episode identifier and step index. Rendering is at 224×224 to match the existing world-model preprocessing. Run Gate 0 and fix the environment.

Stage 1: representation probe (gating). Train linear and small MLP probes from latents to quantities that predict a fall (torso height, pitch angle, angular velocity), on encoder latents and on predicted latents at rollout depths $1 \dots H$. Report R^2 and the degradation curve. Compare substrates. **Kill test:** if the fall-predictive quantity is not decodable at rollout depth, no rollout-based irreversibility estimate in that latent can work, and the project reports that as the finding and falls back to the probe subspace.

Stage 2: irreversibility estimation. Two estimators developed in parallel. The *rollout* estimator samples K action sequences from a fixed distribution, rolls them forward through the predictor, and scores $\rho = \min_j d(\hat{z}_j, z_t)$; fixing the action distribution is what makes differences reflect dynamics rather than remaining control authority. The *precedence* estimator (Grinsztajn et al. 2021) trains a classifier on unlabelled trajectories to predict which of two latents came first, and treats confident asymmetry as a marker of irreversibility; it requires no rollout at inference. Forward dispersion is implemented as the comparison expected to lose. Evaluation is by AUROC against the simulator’s termination flag, held out and never used in training.

Two controls are mandatory and are designed before the experiment is run. States with reduced control authority but no damage must not score as irreversible. The rollout horizon H must be swept to establish that the score does not simply track predictor drift.

Stage 3: supervised baseline and transfer. Train a latent safety filter in the UNISafe style (Seo, Nakamura, and Bajcsy 2025) on Mode A labels only. Train the proposed method on the identical trajectories with no labels. Evaluate both on Mode B. Report dataset composition first: the shared unlabelled set must contain Mode B instances or the result is void.

Stage 4: the dial. Sweep the threshold at deployment on a single trained model. Report irreversible-failure rate against task return as the threshold varies, with conformal calibration supplying a coverage statement, and compare the compute required to trace the whole frontier against penalty-swept baselines that need one training run per point.

5.1 Substrates

Three, which together form the representation axis of H1.

- **Frozen general-purpose features with a small action-conditioned predictor**, in the DINO-WM style (Zhou et al. 2024). Only the predictor is trained, which is cheap and carries little risk, and the encoder discards nothing task-specific. This is the workhorse.
- **An end-to-end JEPA** trained on the selected environment, in the LeWM style (Maes et al. 2026), whose isotropic latent follows from SIGReg (Balestriero and LeCun 2025). This is the substrate H1 predicts will lose.
- **Released LeWM checkpoints on Push-T**, retained as the sanity substrate where the existing pipeline is known to work.

World-model training is infrastructure, not contribution. The substrate comparison is deliberately not load-bearing: if the end-to-end JEPA cannot be trained to a usable standard in the time available, H1 is reported on the frozen-feature substrate alone and the comparison is dropped.

6 Evaluation and baselines

Safety-Gymnasium supplies the baselines that make the result legible: PPO-Lagrangian, CPO (Achiam et al. 2017), TRPO-Lagrangian and related constrained methods, together with Safe-Dreamer (Huang et al. 2024) as the model-based comparison. The supervised latent safety filter of Stage 3 is the baseline the central claim is measured against.

Primary metrics: AUROC of the irreversibility score against held-out termination; Mode B failure rate under each filter; task return; and compute to trace the safety-performance frontier. The penalty-sweep frontier already measured in pilot work (Section 7) is the comparison for the last of these.

7 Preliminary evidence

Pilot work on Push-T with a released LeWM checkpoint (192-dimensional latents) predates this revision but establishes two things the plan depends on.

Physical quantities are decodable from JEPA latents, including imagined ones.

Probes from LeWM latents to pusher position over 100k frames reach $R^2 = 0.965$ (ridge, RMSE 18.9 px) and $R^2 = 0.999$ (MLP, RMSE 3.2 px). On *predicted* latents at rollout depth five the MLP probe still reaches $R^2 = 0.98$, with radial error of median 10.2 px, 95th percentile 35.5 px and maximum 112 px. Physical state survives the predictor, which is what Stage 1 generalises to fall-predictive quantities.

Penalty-based safety fails in exactly the way the dial is meant to address.

A penalty-augmented CEM planner was swept over seven penalty weights and five seeds against a calibrated hazard region. At zero penalty the planner violated on 27% of steps and finished 16.4 px from the goal. At every non-zero penalty the final block error was 70.7 px, which is precisely the start-to-goal distance, meaning the block never moved at all, and the pusher left the arena in every run. Safety was bought by abandoning the task at every operating point that was not zero. This is a first-hand instance of the fragility that motivates moving the operating point out of training and into deployment.

8 Risks and limitations

- **Mode disjointness may not survive trained policies.** This is the largest risk and Gate 0 exists to detect it early. The Hopper result above already shows the modes can co-occur under an aggressive policy.
- **Irreversibility is not harm.** Task progress can be irreversible. The scope is limited to environments whose harmful failures are the irreversible ones, and this is stated rather than defended.
- **Horizon-bounded.** What is measured is “cannot return within H ”, not “cannot return ever”.
- **Predictor reliability near the boundary.** Rollouts close to failure are where the predictor is least trustworthy, which may require an uncertainty gate in the style of UNISafe.
- **Probe supervision.** Probes are supervised on generic physical quantities, never on the safety concept. This is stated explicitly wherever probe results are reported, and the probe-space variant is a fallback, not the headline.
- **Simulation only.** The irreversibility argument is strongest as a claim about physical systems, and no physical validation is in scope.
- **Preprint rate.** The area is moving quickly. Novelty is rechecked against arXiv and OpenReview before submission.

9 Timeline and infrastructure

Period	Work
Late September 2026	Stage 0 harness, Gate 0, environment fixed
October 2026	Stage 1 probes and gating decision; Stage 2 estimators begun
Early November 2026	Stage 2 AUROC with both controls
November 2026	Thesis write-up
December 2026	Stage 3 supervised baseline and transfer
January 2027	Stage 4 dial and frontier; results frozen mid-January
1 February 2027	ICLR 2027 workshop submission

The accepted ICLR 2027 workshop list is announced on 29 November 2026; individual calls cluster in late January and early February. Workshops run 29 to 30 April 2027 in San Francisco. Short-paper tracks of three to five pages are mandatory at every ICLR workshop, with full tracks typically four to nine pages.

Infrastructure note. Safety-Gymnasium 1.0.0 is pinned to Python ≤ 3.10 , gymnasium 0.29 and numpy < 2 , while the existing world-model stack requires Python 3.11, gymnasium 1.3 and numpy 2.x. These cannot share an environment. The intended arrangement is two environments with an offline handoff: trajectory collection in the Safety-Gymnasium environment, written to HDF5; world-model training and evaluation in the main environment. Headless rendering is confirmed working through EGL at the required resolution.

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